EABAA
AECBA
EDCCA
CADCA
ACEEE
BEAD

4.5 oligopoly

1. When two firms interact in an oligopolistic market, which of the following statements is true?

(A) If one firm has a dominant strategy, then the other firm does not have a dominant strategy.
(B) If one firm has a dominant strategy, then the other firm also has a dominant strategy.
(C) Both firms must have dominant strategies.
(D) If one firm has a dominant strategy, then there is no Nash equilibrium.
(E) If both firms have dominant strategies, then there is a Nash equilibrium. ✓

Answer E

Correct. If both firms have a dominant strategy, there will necessarily be a Nash equilibrium.

2. If the three largest widget producers control 85 percent of the total widget market, then these producers are operating in

(A) an oligopoly ✓
(B) monopolistic competition
(C) perfect competition
(D) a monopoly
(E) a cartel

3. A cartel is difficult to maintain for which of the following reasons?

(A) Consumers substitute away from the good when the price increases.
(B) Individual cartel members are tempted to cheat on the agreement. ✓
(C) Although the total gain to cartel members is positive, all members lose when everyone sticks to the agreement.
(D) Some firms will reduce output in an effort to lower costs of production.
(E) Oligopolistic behavior is generally predictable.

4. A collusive agreement to fix prices among firms in an oligopolistic industry is most likely to be broken under which of the following conditions?

(A) It is easy for new firms to enter into the industry. ✓
(B) All of the firms have identical costs.
(C) The number of firms is few.
(D) Firms' sales are widely reported.
(E) The market demand is stable.

4.5 oligopoly

5. Assume that Alpha and Beta are the only sellers of a product and they do not cooperate. Each firm has to decide whether to raise the product price. The payoff matrix below gives the profits, in dollars, associated with each pair of pricing strategies. The first entry in each cell shows the profits to Alpha, and the second, the profits to Beta.

		Beta	
		Raise Price	Do Not Raise Price
Alpha	Raise Price	$100, $100	$30, $120
	Do Not Raise Price	$110, $20	$50, $70

Assuming both firms know the information in the matrix, which of the following correctly describes the dominant strategy of each firm?

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(A)

Alpha	Beta
Do not raise price	Do not raise price

(B)

Alpha	Beta
Do not raise price	Raise price

(C)

Alpha	Beta
Raise price	No dominant strategy

(D)

Alpha	Beta
Raise price	Do not raise price

(E)

Alpha	Beta
No dominant Strategy	Raise price

4.5 oligopoly

6.

PAM'S PRICE POLICY	TARA'S PRICE POLICY: High	TARA'S PRICE POLICY: Low
High	$40, $40	$20, $70
Low	$70, $20	$30, $30

Pam and Tara run two competing lemonade stands in a town. In the payoff matrix above, the first entry in each cell shows the profits to Pam, and the second entry in each cell shows the profits to Tara. According to the information, which of the following is true?

(A) If Pam sets the high price, Tara will do best by charging the low price. ✓

(B) If Pam sets the low price, Tara will do best by charging the high price.

(C) The dominant strategy for both is to charge the high price.

(D) The dominant strategy for Tara is to set the low price and for Pam is to set the high price.

(E) Neither Tara nor Pam has a dominant strategy.

7.

Bright	Sparkle: Strategy 1	Sparkle: Strategy 2
Strategy 1	$3,000, $6,000	$2,000, $4,000
Strategy 2	$500, $3,000	$2,500, $0

The payoff matrix above shows the profits associated with the strategic decisions of two oligopoly firms, Bright Company and Sparkle Company. The first entries in each cell show the profits to Bright and the second the profits to Sparkle. What are the dominant strategies for Bright and Sparkle, respectively?

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(A)

Bright	Sparkle
Strategy 1	Strategy 1

(B)

Bright	Sparkle
Strategy 1	Strategy 2

(C)

Bright	Sparkle
Strategy 2	Strategy 1

(D)

Bright	Sparkle
Strategy 2	No dominant strategy

(E) ✓

Bright	Sparkle
No dominant strategy	Strategy 1

4.5 oligopoly

8.

		Nature View: Bids High	Nature View: Bids Low
Evergreen	Bids High	$600, $300	$520, $400
	Bids Low	$720, $100	$500, $200

Evergreen and Nature View are bidding for a landscaping contract. The payoff matrix above shows what each firm's total weekly profits from all its operations will be for each combination of bids. The first entry in each cell shows Evergreen's profit, and the second entry in each cell shows Nature View's profit. A Nash equilibrium results under which of the following conditions?

(A) When Evergreen bids low, no matter what Nature View's bid is

(B) When both firms bid high and when both firms bid low

(C) When Evergreen bids high and Nature View bids low ✓

(D) When both firms bid low

(E) When both firms bid high

Question refer to the table below.

The two firms in an industry are deciding whether to advertise. The profit to each firm depends on the other firm's decision. The first entries in the matrix below indicate the profit earned, in millions of dollars, by Firm A; and the second entries indicate the profits earned, in millions of dollars, by Firm B.

		Firm B: Advertise	Firm B: Do Not Advertise
Firm A	Advertise	$7, $1	$5, $4
	Do Not Advertise	$3, $2	$2, $0

9. Based on the payoff matrix, which of the following is correct?

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(A) Firm A always gets a smaller share of the industry profits.

(B) Firm A's dominant strategy is to advertise. ✓

(C) Firm B's dominant strategy is not to advertise.

(D) The dominant strategy for both firms is not to advertise.

(E) Neither firm has a dominant strategy.

10. The combination where Firm A advertises and Firm B does not advertise is Nash equilibrium because

(A) it is best for each firm given what the other firm has chosen ✓

(B) the total industry profits are maximized

(C) Firm A has an incentive to change its strategy and chooses not to advertise

(D) It is the best outcome for Firm B regardless of what firm A does

(E) advertising is always the best strategy for Firm A

---

11. Suppose that the two biggest producers of gold, Bmine and Gmine, form a cartel to set price. However, each has the option to cheat or to not cheat on the agreement. The table below shows the payoffs from these strategies, with the first entry in each cell representing the payoff to Bmine and the second representing the payoff to Gmine.

		Gmine	
		Cheat	Not Cheat
Bmine	Cheat	\$10, \$5	\$25, \$20
	Not Cheat	\$5, \$15	\$20, \$25

Which of the following correctly describes the dominant strategy of each firm?

(A) Neither Gmine nor Bmine has a dominant strategy.

(B) Gmine's dominant strategy is to not cheat; Bmine does not have a dominant strategy.

(C) Gmine's dominant strategy is to cheat; Bmine does not have a dominant strategy.

(D) Gmine's dominant strategy is to cheat; Bmine's dominant strategy is to not cheat.

(E) Gmine's dominant strategy is to not cheat; Bmine's dominant strategy is to cheat. ✓

12. Which of the following best describes firms in an industry if all the firms are in a cartel?

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(A) They produce the allocatively efficient quantity.
(B) They face a perfectly elastic demand curve.
(C) They have identical cost curves.
(D) They coordinate their production decisions. ✓
(E) They agree to remove barriers to entry.

Answer D

Correct. Cartels occur when firms that previously competed against each other collude by forming a cartel to coordinate their production decisions.

13. If the only two firms in an industry successfully collude to maximize their joint profit, the price for the product will be

(A) equal to the marginal cost of production
(B) equal to the average total cost of production
(C) above the marginal cost of production ✓
(D) above the monopoly price
(E) below the average variable cost of production

14. Collusion, price leadership, and price wars are usually observed in which of the following market structures?

(A) Perfect competition
(B) Monopolistic competition
(C) Oligopoly ✓
(D) Monopoly
(E) Natural monopoly

There are four firms in an oligopolistic industry. The four firms agree to collude and act like a monopoly.

15. If one of the firms violates the agreement and charges a lower price or sells a larger quantity than what was agreed to, what will happen in the short run?

(A) The firm that cheats will earn higher profits, and industry profits will be lower. ✓
(B) The firm that cheats will earn higher profits, and industry profits will be higher.
(C) The firm that cheats will earn lower profits, and industry profits will be lower.
(D) The firm that cheats will earn lower profits, and industry profits will be higher.
(E) The firms that do not cheat will earn higher profits.

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16. The project funding decisions made by two interdependent research companies, Creative and Genezyz, resulted in a profit of $4 million for Creative and a profit of $5 million for Genezyz. Which of the following is required for this combination of project choices to be a Nash equilibrium?

(A) Genezyz charges a lower price to its customers than Creative does.

(B) Neither firm has a dominant strategy.

(C) Neither firm could earn higher profits by unilaterally changing its project choice. ✓

(D) Creative could earn a higher profit if it chose a different project when Genezyz chose its initial project.

(E) The combined profits of the two firms are maximized at the current combination of project choices.

Answer C

Correct. A Nash equilibrium exists when neither firm can unilaterally increase its profits by altering their strategy.

17. In a market with two firms, a firm that has a dominant strategy will do which of the following?

(A) Maintain that strategy independent of the strategies chosen by its competitor. ✓

(B) Adjust its strategies based on the strategies chosen by its competitor.

(C) Make the first move, and wait to see whether its competitor responds to the move.

(D) Keep its competitor guessing about its next move.

(E) Collude with its competitor to reduce uncertainty.

18.

		Zeb's	
		Lower Prices	Same Prices
Art's	Lower Prices	$300 ; $500	$200 ; $600
	Same Prices	$100 ; $700	$400 ; $800

The above payoff matrix illustrates the daily profits for two restaurants. Each restaurant has the choice to lower prices for early bird customers or keep prices the same. The first entry in each cell indicates the profits for Art's, and the second entry in each cell indicates the profits for Zeb's. Each restaurant independently and simultaneously chooses its action and has complete information of the payoff matrix. Based on the information, does either firm have a dominant strategy?

(A) The dominant strategy for Art's is to lower prices.

(B) The dominant strategy for Art's is to charge the same prices.

(C) The dominant strategy for Zeb's is to lower prices.

(D) The dominant strategy for Zeb's is to charge the same prices. ✓

(E) Neither restaurant has a dominant strategy.

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Answer D

Correct. Independent of Art's choice, the profit for Zeb's is always greatest when Zeb's charges the same price.

		Beta	
		Price High	Price Low
Alpha	Price High	\$150;\$150	\$120;\$180
	Price Low	\$180;\$120	\$125;\$125

The payoff matrix above shows the profits of two firms, Alpha and Beta, that compete against each other. Each firm must decide to set a high or low price. The first numeric entry shows Alpha's profits; the second entry shows Beta's profits. Each firm is aware of the information in this payoff matrix.

19. Given that each firm is aware of the information in the payoff matrix, which of the following is true?

(A) Neither Alpha nor Beta has a dominant strategy.

(B) Both Alpha and Beta have a dominant strategy to price high.

(C) Both Alpha and Beta have a dominant strategy to price low. ✓

(D) Alpha has a dominant strategy to price low, whereas Beta has a dominant strategy to price high.

(E) Alpha has a dominant strategy to price high, whereas Beta has no dominant strategy.

Answer C

Correct. Each firm has a dominant strategy to price low. If Beta priced high, Alpha would earn \$150 if it priced high but would earn more, \$180, if it priced low. So, Alpha would price low when Beta prices high. If Beta priced low, Alpha would earn \$120 if it priced high and \$125 if it priced low. So, Alpha would price low when Beta prices low. Therefore, Alpha would price low regardless of what Beta charges. Beta also has a dominant strategy to price low. Beta would earn more profits charging a low price regardless of the price Alpha charges.

20. Nash equilibrium occurs with which combination of strategies?

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(A) Both firms charging a low price ✓
(B) Both firms charging a high price
(C) Alpha charging a low price and Beta charging a high price
(D) Alpha charging a high price and Beta charging a low price
(E) There is no equilibrium in this game

Answer A

Correct. Nash equilibrium occurs when both firms follow their dominant strategy to charge a low price; each firm will earn $125. It is a Nash equilibrium because neither firm has an incentive to change its strategy since if they do so they each will end up earning lower profits. Nash equilibrium is a noncooperative outcome.

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21. A player's dominant strategy is

(A) independent of the other player's choice ✓
(B) dependent on the other player's choice
(C) the strategy that results in the maximum combined payoff for both players
(D) the strategy that always delivers the other player's best outcome of the game
(E) the strategy that always delivers the other player's worst outcome of the game

Answer A

Correct. A player's dominant strategy is defined as the strategy that maximizes a player's payoff regardless (independent) of the other player's choice.

4.5 oligopoly

		Town Herald	
		Do Not Change Subscription Price	Increase Subscription Price
Daily Voice	Do Not Change Subscription Price	$500, $400	$200, $700
	Increase Subscription Price	$600, $300	$300, $100

The two major newspapers in a city, *Daily Voice* and *Town Herald*, are considering whether to raise the subscription price. The first entries in the matrix above show the profits to *Daily Voice*, and the second entries show the profits to *Town Herald*.

22. Which of the following is consistent with the payoff matrix?

(A) Do Not Change Subscription Price is a dominant strategy for *Daily Voice*.

(B) Do Not Change Subscription Price is a dominant strategy for *Town Herald*.

(C) Increase Subscription Price is a dominant strategy for *Daily Voice*. ✓

(D) Increase Subscription Price is a dominant strategy for *Town Herald*.

(E) There are no dominant strategies in the above payoff matrix.

Answer C

Correct. If *Town Herald* selects Do Not Change Subscription Price, *Daily Voice* does best by selecting to Increase Subscription Price ($\$600 > \$500$). If *Town Herald* chooses to Increase Subscription Price, *Daily Voice* does best by also selecting to Increase Subscription Price ($\$300 > \$200$). Therefore, Increase Subscription Price is a dominant strategy for *Daily Voice*.

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23. Downtown Tech Company and Uptown Tech Company compete in the gaming console market. Downtown Tech Company is deciding whether to charge a high price or a low price for its product. Uptown Tech Company is deciding whether to invest in research and development or advertising. Each company's profit depends on the actions taken by the other firm, which are listed in the payoff matrix provided. The first entry in the matrix is Downtown Tech Company's profit, and the second is Uptown Tech Company's profit. Each firm independently and simultaneously selects an action. Each firm knows all of the information about the payoffs associated with the strategies each firm can choose.

		Uptown Tech Company	
		Research and Development	Advertising
Downtown Tech Company	High Price	$500, $250	$100, $700
	Low Price	$200, $400	$600, $350

The Nash equilibrium to this game, if one exists, is which of the following?

(A) High Price, Research and Development

(B) Low Price, Advertising

(C) High Price, Advertising

(D) Low Price, Research and Development

(E) None of the strategy pairs represents a Nash equilibrium. ✓

Answer E

Correct. Neither firm has a dominant strategy, therefore, none of these strategies represents a Nash equilibrium.

4.5 oligopoly

24. E Soda and R Soda are the only two firms in the soft-drink industry. The companies cannot cooperate. Each firm can follow a high-price strategy or a low-price strategy for pricing its product. In the payoff matrix below, the first entry in each cell shows the profits to E Soda and the second entry shows the profits to R Soda.

		R Soda: Low Price	R Soda: High Price
E Soda	Low Price	$25, $25	$90, $15
E Soda	High Price	$15, $90	$75, $75

Given the information in the payoff matrix, it can be concluded that

(A) neither E Soda nor R Soda has a dominant strategy

(B) E Soda has a dominant strategy but R Soda does not

(C) R Soda has a dominant strategy but E Soda does not

(D) both firms will choose the high-price strategy

(E) both firms will choose the low-price strategy ✓

25. One difference between oligopolies and monopolistically competitive markets is that

(A) there is no deadweight loss in monopolistically competitive markets, but there is in oligopolies

(B) the products sold in monopolistically competitive markets are identical

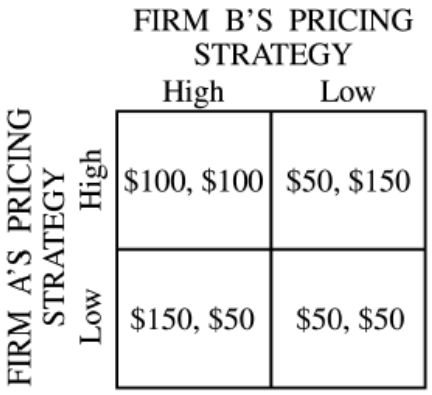
(C) oligopolies have fewer barriers to entry

(D) firms maximize profits in monopolistically competitive markets but not in oligopolies

(E) there are fewer firms in oligopolistic markets than in monopolistically competitive ones ✓

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The following questions are based on the following matrix.



The payoff matrix below gives the profits associated with the strategic choices of two firms in an oligopolistic industry. The first entry in each cell is the profit to Firm A and the second to Firm B.

26. If the two firms collude, Firm A's and Firm B's profits would be which of the following?

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(A)

Firm A	Firm B
$150	$ 50

(B) ✓

Firm A	Firm B
$100	$100

(C)

Firm A	Firm B
$100	$150

(D)

Firm A	Firm B
$ 50	$100

(E)

Firm A	Firm B
$ 50	$ 50

27. If each firm simultaneously chooses its pricing strategy without collusion, Firm A's and Firm B's profits would be which of the following?

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(A)

Firm A's Profit	Firm B's Profit
$150	$ 50

(B)

Firm A's Profit	Firm B's Profit
$100	$100

(C)

Firm A's Profit	Firm B's Profit
$100	$150

(D)

Firm A's Profit	Firm B's Profit
$ 50	$100

(E) ✓

Firm A's Profit	Firm B's Profit
$ 50	$ 50

---

28. In which of the following market structures do firms recognize their mutual interdependence?

(A) Oligopoly ✓
(B) Monopoly
(C) Perfect competition
(D) Unregulated natural monopoly
(E) Monopsony

29. If the four largest firms in a market produce 88 percent of total industry output, the market is

4.5 oligopoly

(A) perfectly competitive
(B) a pure monopoly
(C) a natural monopoly
(D) an oligopoly ✓
(E) a monopsony

Answer D

Correct. In an oligopoly, there are few firms in the market, and they dominate the market. In the scenario provided, the four largest firms produce 88 percent of total industry output, which is consistent with the characteristics of an oligopoly.

30. The following table shows the profits associated with the pricing strategies of two oligopolistic firms, Agronomia and Farmingdale. Each firm has two possible strategies: to charge a low price or a high price. The first entry in each cell shows the profits to Agronomia and the second the profits to Farmingdale.

	Farmingdale charges high price.	Farmingdale charges low price.
Agronomia charges high price.	$150, $150	$50, $300
Agronomia charges low price.	$300, $50	$100, $100

If the two firms do not cooperate, as a result of the firms' pricing decisions the profits of each firm will be which of the following?

4.5 oligopoly

(A)

Agronomia's Profit	Farmingdale's Profit
$50	$100

(B)

Agronomia's Profit	Farmingdale's Profit
$150	$150

(C)

Agronomia's Profit	Farmingdale's Profit
$300	$50

(D) ✓

Agronomia's Profit	Farmingdale's Profit
$100	$100

(E)

Agronomia's Profit	Farmingdale's Profit
$300	$300

31. The use of game theory to explain strategic behavior among firms is most associated with which of the following market structures?

(A) Perfect competition
(B) Monopolistic competition
(C) Oligopoly ✓
(D) Monopoly
(E) Monopsony

32. Game theory is a useful model to explain the behavior of firms in a market when the firms are

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(A) independent of one another
(B) price takers
(C) regulated by government
(D) altruistic
(E) interdependent ✓

33. Game theory is used to explain

(A) why firms price discriminate
(B) how monopolies evolve into oligopolies
(C) strategic behavior of firms in oligopoly ✓
(D) profit maximization in monopoly
(E) price leadership of monopolistic competition

34. Game theory is most commonly used for analyzing the pricing behavior of firms in which market structure?

(A) Perfect competition
(B) Monopolistic competition
(C) Oligopoly ✓
(D) Monopoly
(E) Monopsony

35. Game theory is most useful in describing outcomes in markets where

(A) firms are price takers
(B) there is only one producer
(C) there are many small producers
(D) products are identical for all firms
(E) there are interdependent firms ✓

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36. The table below shows the profits associated with the strategies of two oligopolistic firms, Lock and Star, that must choose between a high price and a low price for their products. The first entry in each box is the profits received by Lock, and the second entry is the profits received by Star.

		Star: High Price		Star: Low Price	
Lock: High Price		\$70	\$70	–\$20	\$50
Lock: Low Price		\$50	–\$20	\$40	\$40

If Lock chooses to charge the low price, the best course of action for Star would be to charge the

(A) low price and earn a profit of \$40 ✓
(B) low price and earn a profit of \$50
(C) high price and earn a profit of \$50
(D) high price and earn a profit of \$70
(E) high price and incur a loss of \$20

37. The cartel model of oligopoly predicts that

(A) all the firms in the industry act in unison to set a monopoly price ✓
(B) each producer acts independently of others
(C) firms follow the low-price firm in the industry
(D) differences in cost of production discourage individual firms from cheating
(E) the markup on marginal cost should be the same for all firms

38. Interdependence among firms is a characteristic primarily associated with

(A) labor markets
(B) perfect competition
(C) monopsony
(D) oligopoly ✓
(E) monopoly

39. In which of the following market structures is it sometimes assumed that rival firms will match price decreases but not match price increases?

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(A) Perfect competition

(B) Oligopoly ✓

(C) Natural monopoly

(D) Monopolistic competition

(E) Monopoly

40.

		JCM	
		Project X	Project Y
RKB	Project Alpha	$300, $400	$250, $250
	Project Beta	$350, $250	$400, $300

RKB and JCM are two competing firms that are considering which project to develop. RKB will develop either Project Alpha or Project Beta. JCM will develop either Project X or Project Y. The payoff matrix provided shows the payoffs for each combination of strategies, and both players (RKB and JCM) have complete information. The first value in each cell represents the payoff for RKB, and the second value represents the payoff for JCM. Each firm independently and simultaneously chooses its strategy.

Which of the following conclusions is supported by the information in the payoff matrix?

(A) There is no combination of project choices that results in a Nash equilibrium.

(B) There are multiple combinations of project choices that result in Nash equilibria.

(C) RKB has a dominant strategy, but JCM does not. ✓

(D) JCM has a dominant strategy, but RKB does not.

(E) Neither firm has a dominant strategy.

Answer C

Correct. RKB has a dominant strategy to pursue Project Beta, as the payoffs to that strategy, regardless of actions taken by JCM, are higher than if it pursues Project Alpha. JCM does not have a dominant strategy, as its payoff to Project X or Project Y is contingent on the strategy pursued by RKB.

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41.

		Zeb	
		Lower Prices	Same Prices
Art	Lower Prices	$300; $400	$600; $200
	Same Prices	$100; $700	$400; $500

The above payoff matrix illustrates the daily profit for two restaurant owners, Art and Zeb. Each owner has the choice to lower prices for early bird customers or keep prices the same. The first entry in each cell indicates the profits for Art, and the second entry in each cell indicates the profits for Zeb. Each restaurant independently and simultaneously chooses its action and has complete information of the payoff matrix. Which of the following is a Nash equilibrium?

(A) Both Art and Zeb will lower prices. ✓

(B) Art will lower prices, and Zeb will charge the same prices.

(C) Art will charge the same prices, and Zeb will lower prices.

(D) Both Art and Zeb will charge the same prices.

(E) There is no Nash equilibrium.

Answer A

Correct. In Nash equilibrium, both Art and Zeb will charge lower prices. The combination of strategies is Nash because neither party has an incentive to change strategy unilaterally to move to any other combination.

North Springs and South Springs are two firms that serve the local market for bottled water. Each firm must choose to either maintain or increase its current output. The first entry in each cell in the payoff matrix below shows the profits for North Springs, and the second entry in each cell shows the profits for South Springs.

		South Springs	
		Maintain Output	Increase Output
North Springs	Maintain Output	$15, $30	$15, $45
	Increase Output	$30, $26	$27, $28

4.5 oligopoly

42. If the two firms do not cooperate, which of the following represents the payoff North Springs and South Springs receive in the dominant-strategy equilibrium and the Nash equilibrium?

(A)

Dominant Strategy	Nash Equilibrium
None	($27, $28)

(B)

Dominant Strategy	Nash Equilibrium
$27, $28)	($15, $45)

(C)

Dominant Strategy	Nash Equilibrium
($30, $26)	($15, $45)

(D) ✓

Dominant Strategy	Nash Equilibrium
($27, $28)	($27, $28)

(E)

Dominant Strategy	Nash Equilibrium
($30, $26)	($27, $28)

4.5 oligopoly

43. The payoff matrix below shows the per-unit profits associated with the production strategies of two utility companies, UA and UB. Each firm has two choices: to reduce production by 10 percent or by 20 percent. The first entry in each cell indicates the profits to UA, and the second, the profits to UB.

		UB	
		Reduces Production by 20 Percent	Reduces Production by 10 Percent
UA	Reduces Production by 20 Percent	$150, $150	$50, $250
	Reduces Production by 10 Percent	$250, $50	$100, $100

Based on the information, and assuming no cooperation, which of the following statements is true?

(A) Neither company has a dominant strategy.

(B) Both companies have an incentive to reduce production by 10%. ✓

(C) Both companies have an incentive to reduce production by 20%.

(D) Only UA has an incentive to reduce production by 20%.

(E) Only UB has an incentive to reduce production by 20%.

44. The characteristic of oligopolistic firms that makes them different from all other types of firms is that oligopolistic firms

(A) are regulated by a state agency or federal agency

(B) consider each other's decisions ✓

(C) advertise their products

(D) produce differentiated products

(E) produce identical products

4.5 oligopoly

45.

		Sam's	
		Lower Prices	Same Prices
Amy's	Lower Prices	$300 ; $400	$600 ; $200
	Same Prices	$100 ; $700	$400 ; $500

The above payoff matrix illustrates the daily profit for two restaurants, Amy's and Sam's. Each restaurant has the choice to lower prices for early bird customers or keep prices the same. The first entry in each cell indicates the profits for Amy's, and the second entry in each cell indicates the profits for Sam's. Each restaurant independently and simultaneously chooses its action and has complete information of the payoff matrix. Based on the information and assuming Amy's and Sam's do not cooperate, which action will each pursue?

(A) Both Amy's and Sam's will lower prices. ✓

(B) Amy's will lower prices, and Sam's will charge the same prices.

(C) Amy's will charge the same prices, and Sam's will lower prices.

(D) Both Amy's and Sam's will charge the same prices.

(E) There is insufficient information to answer the question.

Answer A

Correct. Setting lower prices is a dominant strategy for both Amy's and Sam's.

46. In which of the following market structures is firm interdependence and strategic behavior most commonly observed?

(A) Monopoly protected by a patent

(B) Short-run perfect competition

(C) Monopsony

(D) Oligopoly ✓

(E) Monopolistic competition

Answer D

Correct. In an oligopoly market there are a few firms whose individual actions are predicated on the reaction of rival firms. Thus, interdependence and strategic decision making are prevalent.

47. Which of the following best represents the most distinctive characteristic of an oligopoly?

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(A) A market with a single firm that faces no competition

(B) A market where a few firms have significant market share ✓

(C) A market with low barriers to entry

(D) A market with many firms selling differentiated products

(E) A market with many firms earning zero economic profits in the long run

Answer B

Correct. An oligopoly market is characterized by the existence of only a few interdependent firms, meaning that each firm will have a significant market share of industry revenues.

48. Which of the following best describes an oligopolistic market?

(A) Many sellers with identical products and no barriers to entry

(B) Many sellers, each with a clearly differentiated product, and no barriers to entry

(C) A few competing sellers with similar products and high barriers to entry ✓

(D) A few competing sellers of identical products and no barriers to entry

(E) No competition among sellers and high barriers to entry

49.

		Nature View	
		Bids High	Bids Low
Evergreen	Bids High	$5, $5	$0, $6
	Bids Low	$6, $0	$3, $3

Evergreen and Nature View are bidding for a landscaping contract. The payoff matrix provided shows what each firm's total weekly profit will be for each combination of bids. The first entry in each cell shows the profits of Evergreen, and the second entry shows the profits of Nature View. Evergreen and Nature View act independently and simultaneously, and each firm knows all the information about the payoffs associated with the strategies. Which of the following describes their dominant strategies?

(A) Evergreen does not have a dominant strategy. Nature View's dominant strategy is to bid high.

(B) Evergreen's dominant strategy is to bid high. Nature View does not have a dominant strategy.

(C) Evergreen's dominant strategy is to bid low. Nature View's dominant strategy is to bid high.

(D) Evergreen's dominant strategy is to bid high. Nature View's dominant strategy is to bid high.

(E) Evergreen's dominant strategy is to bid low. Nature View's dominant strategy is to bid low. ✓

4.5 oligopoly

Answer E

Correct. Evergreen's best response is to bid low regardless of Nature View's action. When Nature View bids high, Evergreen will bid low $(\$6 > \$5)$, and when Nature View bids low, Evergreen will also bid low $(\$3 > \$0)$. Therefore, Evergreen's dominant strategy is bid low. Nature View's best response is to bid low regardless of Evergreen's action. When Evergreen bids high, Nature View will bid low $(\$6 > \$5)$, and when Evergreen bids low, Nature View will also bid low $(\$3 > \$0)$. Therefore, Nature View's dominant strategy is bid low.

50. Which of the following best explains why it is difficult to maintain lasting collusive agreements?

(A) There is an unavoidable conflict in that a collusive agreement can increase the profits of some, but not all, firms in the industry.

(B) There is little potential for gain from collusion unless there is a large number of consumers in the market.

(C) Each firm in the industry views itself as facing a vertical demand curve, even though the market demand curve is downward sloping.

(D) The firms in the industry have a common incentive to increase output to a more competitive level.

(E) Each firm realizes that its profits would increase if it were the only firm to violate the collusive agreement by increasing its production slightly. ✓

51. Which of the following characteristics is prevalent in oligopolies?

(A) Allocative efficiency

(B) Low barriers of entry

(C) Consideration of rivals' reactions ✓

(D) No deadweight loss

(E) Zero economic profit

4.5 oligopoly

Rover's Rest and Dave's Den are the only two dog daycare companies in a town. Each firm is considering whether to run ads or to not run ads. The first entry in each cell shows Dave's Den's profit, and the second entry shows Rover's Rest's profit. Each dog daycare company independently and simultaneously chooses its strategy. Assume the two firms know all the information in the matrix and do not cooperate. Use the payoff matrix provided to answer the next two questions.

		Rover's Rest	
		Run Ads	No Ads
Dave's Den	Run Ads	$500, $400	$400, $500
	No Ads	$700, $800	$900, $700

52. Which of the following represents the Nash equilibrium?

(A) Dave's Den chooses "Run Ads," and Rover's Rest chooses "No Ads."

(B) Both Dave's Den and Rover's Rest choose "Run Ads."

(C) Both Dave's Den and Rover's Rest choose "No Ads."

(D) Dave's Den chooses "No Ads," and Rover's Rest chooses "Run Ads." ✓

(E) There are multiple Nash equilibria.

Answer D

Correct. A Nash equilibrium exists when neither firm can unilaterally increase its profits by altering its strategy. When Dave's Den chooses "No Ads" and Rover's Rest chooses "Run Ads," neither firm can alter its strategy to increase its profit.

53. The advertising agency raises its prices, causing a firm that chooses to run ads to earn $200 less in profits. Once the firms adjust to the higher advertising prices, what is the total combined profit of the two firms in the new Nash equilibrium?

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(A) $500
(B) $700
(C) $1,300
(D) $1,500
(E) $1,600 ✓

Answer E

Correct. A Nash equilibrium exists when neither firm can unilaterally increase its profits by altering its strategy. If choosing "Run Ads" causes a firm to earn $200 less in profits, the new Nash equilibrium will be where both Dave's Den follows its dominant strategy of "No Ads" and Rover's Rest follows its new dominant strategy of "No Ads." This will result in the total combined profit of the two firms of $\$900+\$700=\$1,600$.